

ABHISHEK KUMAR

AI Engineer/Consultant

+44-7393068156 | abhishekkumar0598uk@gmail.com | [LinkedIn](#) | Bristol, United Kingdom

PROFESSIONAL EXPERIENCE

AI Engineer/Consultant – Snap Analytics | Bristol, UK | May 2025 – Present

- Proposed and Designed NPS prediction framework for **Jaguar Land Rover (JLR)** using XGBoost and Decision Trees on 550K customers, applying feature engineering pipeline and SHAP explainability to identify loyalty drivers and deploy automated scoring system for targeted retention strategies.
- Own codebase and lead development of Snowflake Native App chatbot for **Edmund Optics** with multi-agent LangGraph architecture, implementing RAG pipeline with embedding models and vector search for semantic retrieval, reducing analyst query load by 35% with <2s response time.
- Proposed and Engineered withdrawal prediction system for **IAM properties** using Snowflake Cortex and Arctic LLM to extract sentiment, pricing and process signals from call transcripts, reducing auction dropouts by 97.5% (400 to 10) via production PowerBI dashboard.
- Delivered inventory optimization solution for **Edmund Optics** using Croston SBA and linear regression with Syntetos Boylan demand classification in Snowpark, replacing \$5,000 licensed software and cutting excess stock by 70% via production Streamlit deployment.
- Proposed and Architected contract intelligence system for **Caffè Nero** using Snowflake Document AI and Arctic models to automate extraction of commercial terms, renewal dates, and SLA clauses, reducing contract review time by 90% with proactive alerts via Streamlit dashboards - production deployed.
- Designed semantic SKU mapping solution for **Premier Foods**, processing 11,000+ SKUs with 92% accuracy using cost-optimized GPT-4o/4o-mini, cutting taxonomy mapping time by 87% (367 to 48) within a large-scale FMCG data integration.

Data Scientist (Finance) – MQube | Manchester, UK | Mar 2024 – Sep 2024 (Affiliated to University)

- Developed production ML pipelines for financial market prediction using AWS SageMaker and Lambda with MLOps best practices, fine-tuned domain-specific LLMs for sentiment analysis and returns forecasting within quantitative trading workflows, improving risk-adjusted returns by 70%.
- Built adaptive quantitative modelling pipelines using rolling statistical windows responsive to real-time financial signals, delivering operational dashboards for trading and risk decision-making.

Data Scientist (Geospatial) – Equans | Manchester, UK | Nov 2023 – Feb 2024 (Affiliated to University)

- Analysed Scope 3 emissions across multi-modal travel data using Databricks, MySQL, and ArcGIS, optimizing routes to reduce CO₂ emissions by 75% and costs by 20%.
- Developed automated ETL pipelines with geospatial analysis (GeoPandas, NetworkX) aligning corporate travel with sustainability goals.

Data Scientist (Fraud and Risk) – National Payments Corporation of India | Nov 2020 – Aug 2023

- Developed 30+ fraud prediction models (Random Forest, Autoencoders, XGBoost) processing 2B monthly UPI transactions, reducing fraud by 35% and false positives by 20%.
- Built end-to-end ML pipelines on Azure using Data Factory, Azure ML, and AKS for production deployment.
- Engineered scalable data ingestion pipelines with Kafka and Azure Event Hubs, supporting high-throughput processing of 2B monthly transactions within a regulated payments environment.
- Performed SQL-based feature extraction and fraud signal engineering on multi-million-row datasets.
- Led A/B testing and designed KPI-driven Power BI dashboards translating ML outputs into business insights

TECHNICAL SKILLS

AI/LLMs: OpenAI, Artic LLM, LangGraph, RAG, Vector DB, Embeddings, Document AI, Prompt Engineering, Fine-tuning, NER, Sentiment Analysis, MCP Servers, Ollama.

Snowflake Ecosystem: Cortex AI, Document AI, Snowpark (UDFs, Stored Procedures, Tasks), Cortex ML, Native Apps, Matillion.

Cloud Platforms: AWS (SageMaker, Glue (ETL), Lambda), Azure (ML Studio, Data Factory, AKS), Databricks, Dataiku.

ML Techniques: Time Series Forecasting, Segmentation, Anomaly Detection, Predictive Modeling, Regression Analysis.

Data Engineering: Fivetran, Kafka, ETL/ELT Pipelines, Azure Event Hubs.

Programming: Python (Pandas, Scikit-learn, TensorFlow, GeoPandas, Shapely, NetworkX), SQL (PostgreSQL), R

MLOps & Automation: Docker, Git, MLFlow, FastAPI

Visualization: Streamlit, Power BI (DAX), ArcGIS, Plotly

EDUCATION

MSc Data Science – Distinction | The University of Manchester, UK | Sep 2023 – Dec 2024

BTech Information Technology – Distinction | SRM Institute of Science and Technology, India | May 2016 – May 2020